MACHINE LEARNING SEMESTER PROJECT

Report Submission Guide

Prepare a research-paper-style project report with clean visuals, captions, and model comparison.

FINAL LAB
9 JULY 2026
Report + PPT + Dashboard + Demo

Important Submission Note

Every group must submit a complete ML project report, create a PPT, build a UI/dashboard, and show the proper working of the project. In the final lab on 9 July 2026, students will explain their project, results, graphs, dashboard, and live demo.

What each group must submit

1. Research-style report

A complete report in research paper format with abstract, introduction, dataset details, methodology, results, discussion, conclusion and references.

2. PowerPoint presentation

A short PPT explaining the project idea, dataset, ML models, graphs, dashboard, model comparison, best model and conclusion.

3. Code / notebook

Submit working Python code, Jupyter Notebook or Google Colab file with proper comments and output screenshots.

4. Dataset file/link

Submit the dataset file or a working dataset link. Mention the source clearly in the report.

5. UI / Dashboard

Create a proper user interface or dashboard that clearly shows project input, workflow, model output/prediction, metrics, and graphs.

6. Proper working demo

The project must work correctly and should be demonstrated in the final lab.

Required report structure

Section	What to write
Title + Authors	Project title, group members, course, semester and submission date.
Abstract + Keywords	Short summary of problem, dataset, models, metrics and best result.
Introduction	Background of the problem and why this ML project is useful.
Problem Statement + Objectives	Clearly define what you are predicting, classifying or analyzing.
Dataset Description	Source, rows, columns, target variable and feature meaning.
Preprocessing	Missing values, encoding, scaling and train-test split.
Models + Methodology	Explain at least 3 ML models and why they were selected.
Results + Discussion	Comparison table, graphs, confusion matrix and explanation of best model.
Conclusion + Future Work	Final result, limitations and how the project can be improved.
References + Appendix	Dataset source, articles/docs, and important code screenshots.

Minimum Standard

- Report must look neat and research-paper style.
- All figures must have titles and captions (Figure 1, Figure 2, etc.).
- All tables must have titles (Table 1, Table 2, etc.).
- Do not paste screenshots without explanation.
- Include dashboard screenshots or interface explanation where relevant.

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Graphs, Model Comparison & Dashboard

Make the report and dashboard visual-friendly using clear charts, comparison tables, and easy explanations.

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Required visuals in the report

1. Target / class distribution
Bar or pie chart showing class balance.

2. Feature distribution
Histogram, box plot or bar chart showing important input features.

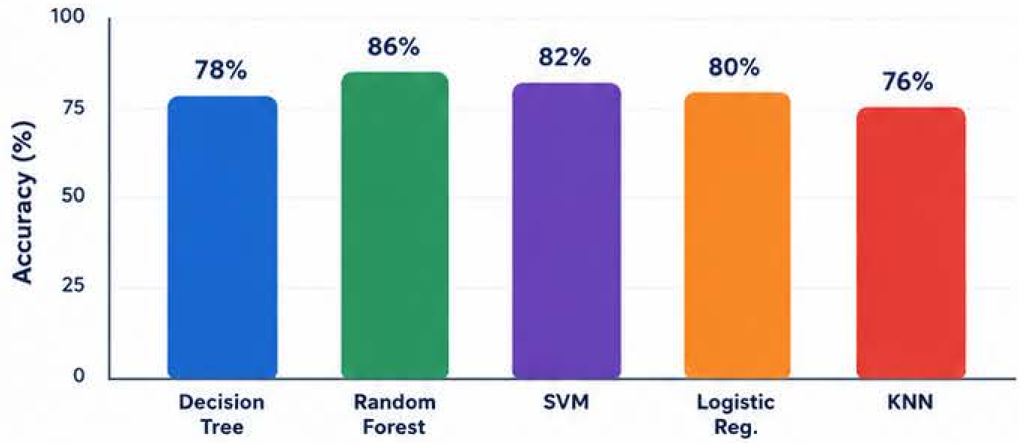
3. Correlation / relationship graph
Heatmap or scatter plot showing feature-target relation.

4. Accuracy comparison
Bar chart comparing trained ML models.

5. Precision / recall / F1
Grouped bar chart comparing evaluation metrics.

6. Confusion matrix
Heatmap showing correct and wrong predictions for the best model.

Example: model comparison graph



What to explain:

- Which model has highest accuracy?
- Is accuracy enough for this dataset?
- Which model has better F1-score?
- Why did the best model perform better?
- What mistakes appear in confusion matrix?

Required model comparison table

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	— %	— %	— %	— %
Random Forest	— %	— %	— %	— %
SVM / KNN / Other	— %	— %	— %	— %
Best Model	Write name	Write reason	Write strength	Final choice

Dashboard note
The UI/dashboard must clearly show the input section, model output or prediction, key metrics, and at least one useful graph or chart.

Caption Rule
Every graph must have a figure number, title, and 2–3 line explanation. If a dashboard is used, include 1–2 screenshots in the report or PPT with short explanation.

MACHINE LEARNING SEMESTER PROJECT

PPT, Dashboard & Final Lab Presentation

Every group must prepare slides, build a dashboard, and explain the complete project during the final lab.

FINAL LAB
9 JULY 2026
Report + PPT + Dashboard + Demo

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Final Lab / Project Presentation

Date: 9 July 2026

Bring your report, PPT, code/notebook and dataset. Each group must explain the project clearly, show the UI/dashboard, and demonstrate the proper working of the system.

PPT slide structure

Slide 1	Title, group members, course and project topic
Slide 2	Problem statement: what problem are you solving?
Slide 3	Dataset: source, rows, columns, target variable
Slide 4	Preprocessing: missing values, encoding, scaling, split
Slide 5	Models used: at least 3 ML algorithms
Slide 6	Graphs: dataset visuals and feature understanding
Slide 7	Results: accuracy, precision, recall, F1-score table
Slide 8	Model comparison graph and best model explanation
Slide 9	UI / dashboard screenshots or demo flow
Slide 10	Conclusion, limitations, future work, and working demo summary

Presentation requirements

- Each group member should explain at least one part of the project.
- Students must explain the dataset, preprocessing, selected models and results.
- Students must show model comparison using graphs, not only text.
- Students should explain why the best model was selected.
- Students must show the UI/dashboard.
- Students must demonstrate the proper working of the project, not only screenshots.
- Students must be ready to answer questions about code, metrics, graphs, dashboard, and project workflow.

Final submission checklist

- ☑ Research-style report
- ☑ Python code / notebook
- ☑ Model comparison table
- ☑ UI dashboard
- ☑ Conclusion + references
- ☑ PPT presentation
- ☑ Dataset file or link
- ☑ Required graphs with captions
- ☑ Proper working demo

Your final marks will depend on report quality, ML pipeline correctness, visual graph comparison, dashboard quality, project working demo, PPT explanation, and your ability to answer project questions.